

EECS-841

Image Proceeding and Computer Vision Midterm 1 Exam, Session: Spring'25

Student's name:

ID:

Date:

Exam Guidelines:

1. You have 60 minutes to complete this exam.
2. The exam is closed book. No cheat sheet is allowed.
3. Don't keep any book, cell phone, tab, laptop, copies, or notebook at your desk while you are attending the exam.
4. Detailed work/explanation on problems is suggested to receive partial credit.
5. Complete all your answers within the given answer sheet. No additional page will be provided.
6. The exam is meant to test your understanding. So be patient and read the questions/problems carefully before you answer.
7. If you don't submit your copy on time/when asked, you will get a penalty of 5 pts reduction for every additional minute delay.
8. It is not advised to leave the classroom during exam.

DO NOT do anything that might be perceived as cheating, such as:

- Sharing solutions with each other.
- Getting help from anyone other than yourself.
- If any type of cheating is detected, the minimum penalty is getting zero for the exam. There are more academic consequences because this misconduct will be reported to the department chair.

Scoring:

Question No.	Points allocated	Points obtained by the student
1	20	
2	15	
3	15	
4	20	
5	30	

Total = 100 points

Total points obtained by student:

Assumptions for ALL questions:

- If I is an image, $I(x,y)$ denotes the pixel intensity at (x,y) location where x denotes the column index and y denotes the row index.
- The index starts from 0 for both the column and row.

Question 1. Image A follows the conventional 8-bit pixel representation method. You wish to transform it for a resource-constrained edge device using the n -bit quantization method. What will the output image be if you apply 3-bit quantization to it? (20 points)

188	91	162	11
130	57	112	111
103	187	25	40
220	20	220	203

Image A

Answer:

Question 2. Write the pseudo algorithm for a function called “modify_image” which accepts a grayscale image of size $m \times n$ and returns the output image of the same size meeting the below criteria: (15 points)

- i) The brightest pixel(s) become(s) the darkest pixel(s)
- ii) The darkest pixel(s) become(s) the brightest pixel(s)
- iii) Any pixel(s) with intensity between 175 to 200 (including these pixels themselves) become(s) 120.

Answer:

Question 3. Does the quality of segmentation impact the accuracy of classification results? Please explain with an example. *(15 points)*

Answer:

Question 4. Consider the following single-channel 2D image matrix:

88	91	62
130	78	112
43	187	5

- i) What is the representation of the image if stored in row-major order? *(10 points)*
- ii) Show the graph representation of the image. *(10 points)*

Question 5. Consider the following image:

0	1	1	3	4
7	2	5	5	7
6	3	2	1	1
1	4	4	2	1

- i) Find the histogram of the image. (5 points)
- ii) Does the image have a high contrast? Why or why not, justify. (5 points)
- iii) Considering the following assumptions, perform histogram equalization on the given image. Show the updated equalized histogram and the output image based on that. (10+ 5+ 5 = 20 points)

Assumptions:

- $K_{Max} = 7$
- While calculating the final output pixel intensity in the equalized histogram, consider rounding to two decimal places (e.g., 0.8035 rounded to two decimal places will be 0.80, or 0.8509 will be 0.86)

Answer:

Extra Page: